Not All Layers Need Tuning: Selective Layer Restoration Recovers Diversity

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Abstract

Post-training improves instruction-following and helpfulness of large language models (LLMs) but often reduces generation diversity, which leads to repetitive outputs in open-ended settings, a phenomenon known as mode collapse. Motivated by evidence that LLM layers play distinct functional roles, we hypothesize that post-training-induced diversity loss is unevenly distributed across layers and that restoring a carefully chosen range of layers to their pre-trained weights can produce a wider range of high-quality responses. To test and operationalize this idea, we design Constrained Random Character (CRC), a proxy task with an explicit validity set and a natural diversity objective. Results on CRC reveal a clear diversity–validity trade-off across restoration ranges and identify configurations that increase diversity with minimal quality loss. Based on these findings, we propose Selective Layer Restoration (SLR), a training-free method that restores selected layers in a post-trained model to their pre-trained weights, yielding a hybrid model with the same architecture and parameter count, incurring no additional inference cost. Across Llama, Qwen and Gemma, SLR increases diversity in creative writing and open-ended question answering with little loss in writing quality or answer accuracy. It also improves GSM8K Pass@k.

1 Introduction

Post-training, including instruction tuning [23, 30] and RLHF [19], is a double-edged sword. It improves output quality (e.g., instruction adherence and helpfulness) in large language models (LLMs), but is also observed to induce mode collapse [11, 20]; the post-trained model assigns disproportionately high probability mass to a narrow range of responses (the "mode") among many valid alternatives. This leads to a *reduced output diversity* and is undesirable in settings where multiple distinct responses are valued (e.g., creative writing, open-ended question answering, and reasoning), thereby limiting their effectiveness in these applications.

Existing approaches to mitigating mode collapse can be grouped into three broad families: *decoding-based* [2, 17, 27], *prompt-based* [6, 26, 38], and *training-based* methods [14, 35]. Each comes with distinct limitations; decoding-based methods are limited in recovering behaviors suppressed during post-training, prompt-based methods increase inference cost, and training-based methods require expensive retraining. In this work, we seek a method that (i) recovers diversity effectively, (ii) is computationally efficient during adaptation/training, and (iii) introduces no additional inference-time cost.

We draw on two empirical observations. First, LLM layers exhibit different functional roles [16, 25, 28] (e.g., factual associations tend to be localized to mid-depth layers [16]). Second, much of the model's knowledge is acquired during pretraining, while post-training primarily steers which

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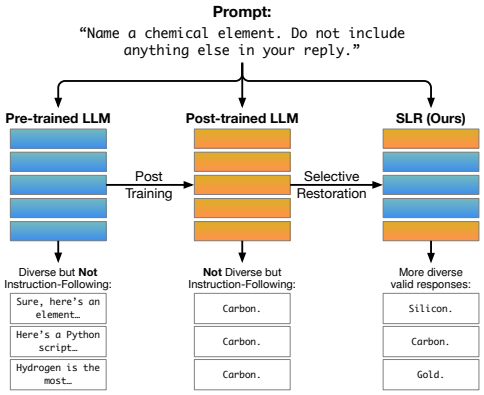


Figure 1: Pre-trained LLMs are diverse but have weak instruction-following ability, leading to low-quality responses. Post-trained LLMs show strong instruction adherence and high output quality but suffer from mode collapse. Selective layer restoration (SLR) increases diversity by restoring selected layers to their pre-trained weights.

pretrained behaviors are expressed and can suppress others (catastrophic forgetting) [12, 13]. Based on these findings, we hypothesize that post-training-induced diversity loss is unevenly distributed across layers, and that **selectively restoring** a carefully chosen set of layers to their **pre-trained parameters** can produce a wider range of high-quality responses.

To test our hypothesis and make it actionable, we reduce it to a concrete selection problem: *which layers should be restored?* We cast this as a *diversity–quality* constrained optimization problem, where the objective is to maximize output diversity while maintaining output quality above a minimum threshold.

This setup entails solving two challenges. As posed, the general problem involves searching over arbitrary subsets of layers, which is a large combinatorial search-space. We make the problem tractable by restoring a *contiguous interval* of layers. A second issue is that directly estimating diversity and quality for each candidate restoration interval on complex downstream tasks (such as creative writing) would be expensive and difficult, especially for "quality" which is often noisy or subjective. To address this, we introduce a simple proxy task, **CRC** (**C**onstrained **R**andom **C**haracter): random digit and letter generation under strict output constraints (e.g., "Generate a random number in the range [0, 5]"). On CRC, both diversity and quality are naturally and unambiguously defined, enabling efficient exploration of restoration intervals and guiding the choice of layers to restore. The proxy serves two purposes: it provides evidence that restoring different intervals induces different diversity–quality trade-offs, and it guides the choice of layers to restore for downstream tasks.

Guided by this proxy, we propose **Selective Layer Restoration (SLR)**, which constructs a hybrid model by restoring a chosen interval of layers in a post-trained LLM to their pre-trained parameters. SLR directly meets our desiderata since it is **training-free** (operating directly on existing checkpoints), **computationally efficient**, and **keeps inference cost unchanged** (it preserves the architecture and parameter count). We evaluate SLR on three downstream tasks that require both diversity and quality: creative writing, open-ended question answering, and multi-step reasoning. We use three model families (Qwen [34], Gemma [7], and Llama [8]). Across these settings, SLR increases diversity in creative writing and open-ended QA with little loss in writing quality or answer accuracy. It also improves GSM8K Pass@k. We also show that SLR is complementary to decoding- and prompt-based methods, yielding additional gains when combined. Finally, we assess CRC's usefulness for interval selection by comparing interval rankings on CRC and open-ended QA.

In summary, our paper makes the following contributions:

- SLR, a simple yet effective training-free weight-space intervention that restores an interval of layers in a post-trained LLM to their pre-trained parameters, without changing the architecture or incurring additional inference cost. It is also complementary to common decoding- and prompt-based diversity interventions.

- CRC, a proxy task with an explicit validity set and natural diversity objective. CRC provides evidence for our hypothesis that post-training-induced diversity loss is unevenly distributed across layers and guides the selection of restoration intervals.
- Empirical evidence that demonstrates the effectiveness of SLR across three downstream tasks and three model families.

More broadly, our findings support a modular view of post-trained LLMs. In particular, post-training does not affect all layers equally, and the diversity loss induced by post-training is distributed unevenly across layers. Our work shows that desirable properties of pre-trained models, like diversity, can be recovered by targeted interventions.

2 Related work

In this section, we summarize related work on mode collapse in post-trained LLMs, existing mitigation strategies, layer-wise functional roles in transformer models, and model merging techniques.

Mode collapse in post-trained LLMs It is widely observed that post-trained LLMs suffer from mode collapse and have significantly lower output diversity compared to their pre-trained counterparts [11, 20, 33]. Mode collapse is often attributed to factors such as KL-regularized optimization used in RLHF [33] and bias in the post-training data [38]. Broadly, post-training can be viewed as steering the pretrained distribution toward responses favored by supervision or preference data, thereby concentrating probability mass on a subset of already-plausible modes and suppressing others [12, 25].

Mitigating mode collapse. Existing methods to mitigate mode collapse can be categorized into three families: (i) Decoding-based methods work by increasing stochasticity (increasing temperature [21]), truncating the distribution by different cutoff thresholds [17, 27], or using heuristics like controlling output perplexity [2]. Since they do not directly alter the model parameters, they are limited in recovering behaviors that may be suppressed during post-training. (ii) Prompt-based methods rely on hand-crafted prompt templates to directly modify the input, e.g., explicitly requesting multiple distinct answers [38], enforcing different perspectives [6]. However, their effectiveness is sensitive to the prompt wording and task context, and the increased prompt length will increase the inference cost. (iii) Training-based methods modify post-training methods to explicitly encourage diversity, e.g., via objective design [14, 35], data collection strategies [24]. Such approaches require re-training and careful tuning, often with non-trivial computational and data cost. In contrast, SLR operates directly in weight space on existing checkpoints, improving diversity without additional training or inference-time overhead.

Functional roles of LLM layers. Prior works [16, 25, 28] suggest that transformer-based LLMs exhibit depth-dependent functional specialization. It is found that earlier layers correlate more with surface/lexical and local syntactic features, and deeper layers more with higher-level semantics [25, 28]. Complementing this, model editing work provides more causal evidence of localization: e.g., ROME and related analyses [16] identify that knowledge storage and retrieval can be localized to mid-depth layers, and modifying them can effectively edit the knowledge stored in the LLMs. Together, these findings motivate our hypothesis that the diversity loss induced by post-training may be distributed unevenly across layers.

Model merging. Model merging combines checkpoints in weight space without changing the architecture or increasing per-token inference cost. A common approach is (optionally weighted) parameter averaging across checkpoints [5, 15, 32, 36], and related work studies other merging operators that combine models at a finer granularity, such as editing or stitching subsets of layers or sub-layer components to transfer or compose capabilities [1, 5, 10, 32, 36]. Although much of this literature focuses on combining multiple post-trained or task-specialized checkpoints ("experts") [5, 32, 36], some recent studies explore merging a post-trained model with its pre-trained counterpart and observe trade-offs between quality and diversity [10]. In our work, SLR performs structured restoration between a post-trained model and its pre-trained counterpart with the explicit goal of improving the diversity–quality trade-off, and uses CRC to guide which components to merge.

3 Methodology

This section presents our primary contribution, SLR, a training-free method for mitigating mode collapse in post-trained LLMs by selectively restoring a contiguous interval of transformer layers to their pre-trained weights.

3.1 SLR: Selective Layer Restoration

Let M_{pre} denote a pre-trained language model and M_{post} its post-trained counterpart, sharing the same transformer architecture with N transformer layers. We write their layer stacks as $M_{\text{pre}} = \{L_0^{\text{pre}}, \ldots, L_{N-1}^{\text{pre}}\}$ and $M_{\text{post}} = \{L_0^{\text{post}}, \ldots, L_{N-1}^{\text{post}}\}$. For any layer interval $0 \le i \le j \le N-1$, we define the hybrid model $M_{i:j}$ by replacing the corresponding post-trained layers with their pre-trained counterparts, i.e., $L_k^{i:j} = L_k^{\text{pre}}$ if $i \le k \le j$, and $L_k^{i:j} = L_k^{\text{post}}$ otherwise.

Given a task $\mathcal{T}$ (e.g., creative writing), we associate each model M with a *quality* score $Q_{\mathcal{T}}(M)$ and a *diversity* score $D_{\mathcal{T}}(M)$. **S**elective **L**ayer **R**estoration (SLR) therefore works by identifying the restoration interval (i, j) that maximizes diversity while maintaining quality above a threshold $q_{\min}$ and constructs the hybrid model accordingly: $\max_{0 \le i \le j \le N-1} D_{\mathcal{T}}(M_{i:j})$ s.t. $Q_{\mathcal{T}}(M_{i:j}) \ge q_{\min}$. A first-order view of why layer restoration can affect output entropy is provided in Appendix A.

3.2 CRC: Constrained Random Character

Directly solving the above problem on downstream tasks, such as creative writing, is challenging: evaluating Q and D requires multiple generations per prompt and expensive task-specific assessment. Instead, we design a simple proxy task, **C**onstrained **R**andom **C**haracter (**CRC**).

CRC is designed so that both diversity and quality are *unambiguous, discrete, and automatically measurable*. Each prompt specifies a finite valid output set and requires *a single-token answer* from that set. In this way, CRC avoids expensive sampling-based evaluation by directly using the model's predicted distribution at the first generation step. We instantiate two families of constraints:

- **Digit constraints**, e.g., "Generate a random integer in the range [0, 5]. Do not include anything else in your reply.", with valid set $\{0, 1, 2, 3, 4, 5\}$ and
- **Letter constraints**, e.g., "Generate a random letter from A to G (inclusive). Do not include anything else in your reply.", with valid set $\{A, B, C, D, E, F, G\}$.

This construction removes semantic ambiguity. For each prompt, the set of valid outputs is explicitly known, and the valid set can be varied, enabling us to generate numerous prompts with different valid sets while using the same evaluation method. In practice, we construct 20 prompts for each constraint family by sampling over valid ranges, e.g., [0,5], [1,7].

For each model family, we format CRC prompts using the default chat template. We encode each admissible character with the post-trained tokenizer using `add_special_tokens=False`; all admissible characters in our experiments map to a single token. The tokenizer and template remain fixed across restoration candidates.

Let D denote the set of prompts. For a prompt $x \in D$ with a valid token set $\mathcal{V}(x)$, let $p_M(\cdot \mid x)$ be the model's next-token distribution at the *first generated position* (i.e., immediately after the input). We define the *quality* score as the probability mass assigned to valid outputs, $Q_x(M) \triangleq \sum_{v \in \mathcal{V}(x)} p_M(v \mid x)$, and the average CRC quality score as $Q_{\text{CRC}}(M) \triangleq \frac{1}{|D|} \sum_{x \in D} Q_x(M)$. This measures the probability of a valid next token, without assessing whether the response stops after that token. Conditioned on producing a valid token, the induced distribution over valid outputs is $\tilde{p}_M(v \mid x) \triangleq \frac{p_M(v|x)}{\sum_{u \in \mathcal{V}(x)} p_M(u|x)}$ for $v \in \mathcal{V}(x)$, and we define the *diversity* score as the entropy over valid outputs $D_x(M) \triangleq -\sum_{v \in \mathcal{V}(x)} \tilde{p}_M(v \mid x) \log \tilde{p}_M(v \mid x)$, and the average CRC diversity score, $D_{\text{CRC}}(M) \triangleq \frac{1}{|\text{D}|} \sum_{x \in \text{D}} D_x(M)$.

Applying CRC using a naïve grid search over all possible interval configurations, i.e., $\frac{N(N+1)}{2}$ intervals given N layers, takes around 3 minutes, 5 minutes, and 20 minutes for Llama, Qwen, and Gemma, respectively, on one single A100 GPU without batching. This search is performed once

per model family; after selection, SLR has the same per-token forward-pass cost as the post-trained model. As such, using CRC as a proxy task to optimize a good interval is computationally low-cost relative to training-based methods (e.g., roughly three days of RL training for a 7B model on eight A6000 GPUs [35]), and future work may look into optimizing this search.

3.3 CRC proxy analysis

Figure 2 summarizes the CRC proxy trade-off across three models: Llama3.1-8B, Qwen2.5-7B, and Gemma2-9B (restricted to intervals with $Q_{\mathrm{CRC}} \geq 0.9$ for visualization clarity). For each model, we plot the (empirical) Pareto-front of contiguous restoration intervals $[i, j]$ (faded points). The post-trained models (diamonds) sit at the high-quality / lower-diversity end of the proxy landscape, which shows substantial headroom for increasing diversity while remaining within a high-quality regime. Interestingly, none of the post-trained models are on the Pareto-front for the CRC task. We observe a smooth diversity-quality trade-off where intervals with higher mean entropy over the valid set also exhibit lower mean validity.

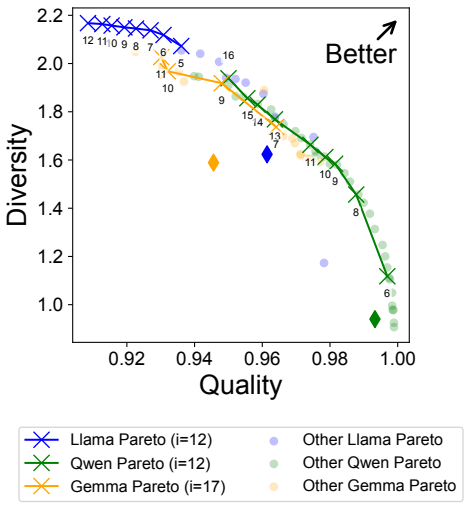


Figure 2: **CRC trade-off landscape.** Each point is a restoration interval $[i, j]$ (filtered to $Q_{\mathrm{CRC}} \geq 0.9$ for visualization) on the Pareto-front, plotted by quality Q_{CRC} (mean validity) and diversity D_{CRC} (mean entropy). The number next to each marker is the number of restored layers $\ell = j - i + 1$. Pareto frontiers are smooth, and among the displayed fixed-start Pareto-optimal intervals, restoring more layers increases diversity at a gradual cost in validity; post-trained models (diamonds $\diamondsuit$) sit at high-validity/low-diversity.

To clarify the relationship between the number of restored layers and the diversity-quality trade-off, we highlight fixed-start slices for visualization (Llama/Qwen: i=12; Gemma: i=17) and connect the corresponding Pareto-optimal intervals in increasing end index j. Among the displayed Pareto-optimal intervals with a fixed start layer, restoring more layers tends to increase diversity and reduce validity.

We instantiate SLR's selection objective as defined in Sec 3.1 using CRC. Let $\mathcal{C} = \{M_{\mathrm{post}}\} \cup \{M_{i:j} : 0 \leq i \leq j < N\}$ denote the candidate set, including post-trained model. For each model family, we set the quality threshold relative to the best CRC quality among candidates: $q_{\min} := 0.95 \cdot \max_{M \in \mathcal{C}} Q_{\mathrm{CRC}}(M)$. We then select the feasible restored interval with the highest $D_{\mathrm{CRC}}(M_{i:j})$.

This yields restoration intervals of $[12, 17]$ for Llama, $[12, 27]$ for Qwen, and $[17, 27]$ for Gemma, which we use in all downstream experiments. At the default threshold, these intervals restore 18.8%, 57.1%, and 26.2% of layers in Llama, Qwen, and Gemma, respectively. Here, "selective" refers to CRC-guided interval selection; the restored fraction varies across model families. In our threshold sweep, lower thresholds select larger intervals and increase open-ended QA entropy and coverage at the cost of precision (Appendix C.11).

4 Experiments

Here, we conduct experiments to answer the following research questions: Q1: Can SLR improve output diversity with minimal loss of quality? Q2: Does SLR generalize across model families? Q3: Does CRC-guided interval selection transfer to downstream tasks? Q4: Is SLR complementary to decoding and prompt-based methods?

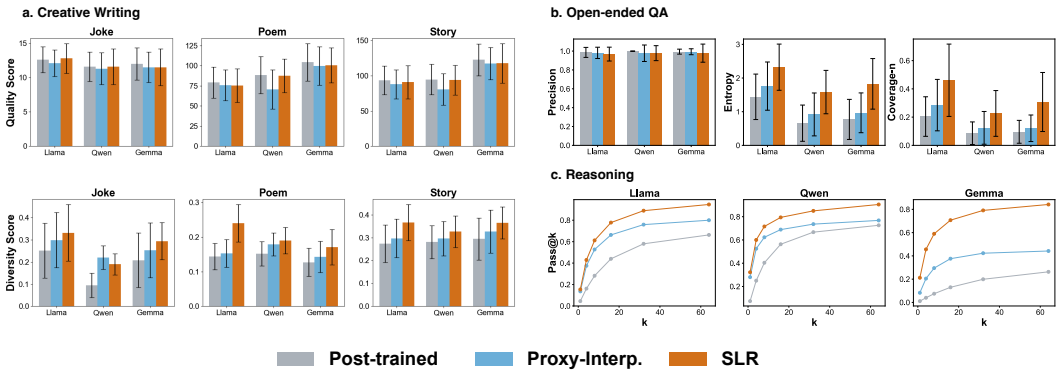


Figure 3: **Main experiment results.** We compare the post-trained model, Proxy-Interpolation, and SLR across Llama, Qwen, and Gemma on creative writing, open-ended QA, and reasoning. (a) **Creative writing results**: top row shows quality (LLM-judge score) and bottom row shows diversity (embedding-based dissimilarity) (b) **Open-ended QA results**: quality is measured by precision, while diversity is measured by the entropy over correct answers and Coverage-n (fraction of unique correct answers generated). (c) **Reasoning results**: Pass@k as a function of the sampling budget k. Overall, the CRC-selected SLR models improve diversity with minimal quality loss and yield higher Pass@k across model families, *providing affirmative evidence for Q1–Q3.*

4.1 Experimental setup

Tasks. We choose creative writing, open-ended QA, and GSM8K to assess variation in content, coverage of correct factual answers, and the chance of sampling a correct solution, respectively.

- Creative Writing: We follow Zhang et al. [38] and evaluate on three benchmarks: poem, story, and joke generation. We use the same datasets as Zhang et al. [38], where each benchmark contains 100 prompts. For each prompt, we sample 32 generations.
- Open-ended QA: We use the adapted CoverageQA [31] benchmark provided by Zhang et al. [38], which contains 40 open-ended questions with a wide range of valid answers (e.g., "Name a national park in the United States."). For each prompt, we sample 128 generations.
- Reasoning: We construct a model-family-specific subset of GSM8K [3], a grade school level math reasoning benchmark, consisting of questions that the corresponding M_{post} fails under greedy decoding, thus requiring exploration. For each model family, we sample 100 such questions, and for each question, we sample 64 generations.

SLR models. Our main experiments focus on evaluating SLR across three models from different families: Llama-3.1-8B [8], Qwen-2.5-7B [34], and Gemma-2-9B [7] (we will denote them as Llama, Qwen, and Gemma for simplicity). For each model, we use a *pre-trained* model M_{pre} and a corresponding *post-trained* model M_{post} (same architecture and layer count), and apply SLR by restoring a contiguous interval of transformer layers of M_{post} back to their pretrained parameters. The restoration interval is obtained as discussed in Section 3 ([12, 17] for Llama, [12, 27] for Qwen, and [17, 27] for Gemma). To test the scalability of SLR to larger models, we additionally evaluate SLR on larger models, Gemma2-27B and Qwen2.5-32B, on Open-ended QA, with results reported in Appendix C.6.

Baselines. We compare SLR against the post-trained model, M_{post}, and global interpolation between M_{pre} and M_{post} [10]: $M_\alpha \triangleq \alpha M_{\text{post}} + (1-\alpha)M_{\text{pre}}$. We select α on CRC using the same quality-constrained diversity objective as SLR, searching $\alpha \in \{0.00, 0.05, \ldots, 1.00\}$. We call this baseline *Proxy-Interpolation* (*Proxy-Interp.* in tables and figures). The selected coefficients are $\alpha = 0.85/0.50/0.90$ for Llama, Qwen, and Gemma, respectively. Additional comparisons with Vanilla-Soup ($\alpha = 0.5$) and HMA [15] are reported in Appendix C.5.

We do not include training-based diversity methods as baselines, since they require additional retraining and access to post-training pipelines/data, whereas SLR is a training-free intervention. To isolate the effect of weight-space interventions, we also hold decoding and prompting fixed across

all methods. We separately study composability with representative decoding and prompt-based diversity interventions in Section 4.3.

Decoding and prompting. We use min-p sampling [17], a state-of-the-art sampling method, with $p_{\text{base}} = 0.1$ in the main experiments and hold decoding fixed across methods to isolate the effect of weight-space interventions. In our main experiments, we use temperature $T = 1$ throughout. For prompting, we use the plain instructions from each dataset and format them using the model's default chat template (as provided by the tokenizer), which we keep fixed across all methods; we do not perform additional prompt engineering. We fix all other decoding hyperparameters across methods as well, including maximum generation length and stopping criteria. For more details, please refer to the appendix B.1.

Evaluation metrics. Since quality and diversity manifest differently across settings, we use task-appropriate metrics. Unless otherwise stated, reported values are means over prompts, and error bars or parenthesized values denote standard deviations across prompts.

- Creative Writing: Following Zhang et al. [38], we quantify diversity with $1 - \bar{s}$ where $\bar{s}$ is the mean pairwise cosine similarity of text embeddings (generated using Qwen3-Embedding-8B [39], a state-of-the-art open-source embedding model) across the 32 samples per prompt and averaged over prompts. To evaluate output quality, we use Gemini-2.5-Pro [4] as the judge model, following the same rubrics as Zhang et al. [38]. See the appendix B.2 for the concrete evaluation prompts and rubrics used.
- Open-Ended QA: We notice that the reference answer lists in CoverageQA are not surface-form exhaustive (e.g., they may omit aliases or near-equivalent variants). To robustly evaluate correctness, we use an LLM-based canonicalization step with Gemini-2.5-Pro similar to Zhang and Soh [37]; given the question, the dataset's set of correct answers, and a model-generated response, we ask an LLM judge to map the response to one of the provided correct answers if it is semantically equivalent, and to return None otherwise. Since the valid answer set is finite, we measure diversity using (i) entropy of the distribution over the generated correct answers and (ii) coverage-n, the proportion of unique correct answers generated at least once. We measure quality using precision, i.e., the fraction of generations that can be mapped to any reference answer.
- Reasoning: We report Pass@k, the probability of obtaining at least one correct final answer among k samples, for $k \in \{1, 4, 8, 16, 32, 64\}$. Answer extraction and estimation details are provided in Appendix B.1.

Statistical analysis. We assess significance using two-sided tests with Holm correction at the 0.05 level, primarily comparing SLR with the post-trained model and secondarily with Proxy-Interpolation. For creative writing and open-ended QA, we use Welch's t-tests on prompt-level means and standard deviations, with $n = 100$ per task and $n = 40$, respectively. For these tasks, we apply Holm correction separately to quality and diversity within each experiment, benchmark, and baseline comparison. Appendix C.12 provides the detailed results and the GSM8K tests, with Holm correction across the 18 comparisons against Post-trained.

4.2 Main results and analysis

We present the main results and findings in this section. Full tables, additional analyses, extra baselines, and qualitative examples are provided in Appendix C.

Creative writing. The bar charts in Figure 3.a summarize the quality and diversity scores obtained by SLR on the three creative writing benchmarks against the baselines. The full results can be found in Table 1. Across all three model families and all three benchmarks, *SLR consistently improves diversity at comparable quality*. SLR achieves the highest diversity in eight of the nine model–task pairs; Proxy-Interpolation is more diverse on Qwen jokes. SLR increases diversity by 41.5% on average, with a 2.1% average decrease in judged quality. Across the nine model–task pairs, diversity gains over the post-trained model are significant ($p_{\text{Holm}} \leq 1.43 \times 10^{-5}$), while no significant difference in judged quality is detected ($p_{\text{Holm}} > 0.05$). In contrast, the Proxy-Interpolation baseline typically delivers smaller diversity gains (on average $+22.6\%$) and incurs more noticeable quality degradation (on average -7.1%), most prominently for Qwen on poem and story.

Open-ended QA. The results on Open-ended QA are summarized in the bar charts in Figure 3.b. The full results can be found in Table 2. *SLR substantially increases answer diversity (on average* $+112.7\%$ *in entropy and* $+169.5\%$ *in coverage-n) while preserving high quality (on average* -1.8% *in precision) across all three model families*. The change in precision is minimal relative to the post-trained model, with no significant difference detected ($p_{\text{Holm}} > 0.05$). In contrast, both diversity metrics improve significantly over the post-trained model for all three models ($p_{\text{Holm}} \leq 8.30 \times 10^{-6}$). Table 13 illustrates the gain for Llama: Post-trained repeats Yellowstone variants that count as one answer, while SLR also generates valid alternatives such as Acadia, Yosemite, and Everglades. The Proxy-Interpolation baseline provides only modest improvements in entropy and coverage compared to SLR, particularly for Qwen and Gemma, where SLR more than doubles coverage relative to the post-trained model. The diversity gains also persist at $T = 0.5$ and at model-specific temperatures (Appendix C.2).

Reasoning. The line charts in Figure 3.c show Pass@k on the GSM8K subsets where the post-trained model fails under greedy decoding. Across all three model families, SLR outperforms both baselines at every $k \in \{1, 4, 8, 16, 32, 64\}$, increasing the probability of sampling a correct final answer at a fixed sampling budget. All 18 gains over the post-trained model remain significant after Holm correction (raw $p \leq 1.53 \times 10^{-5}$). Evaluation on a shared random subset shows the same trend across all three model families (Appendix C.7).

Behavioral trade-offs. While writing quality and QA precision remain close to Post-trained, SLR lowers IFEval strict prompt accuracy by 12.5 to 20.1 percentage points across the three model families. Refusal behavior also shifts: Qwen refuses more safe and unsafe prompts; Llama refuses more safe prompts and fewer unsafe prompts; Gemma refuses fewer unsafe prompts. Layer restoration can therefore recover diversity while partially reversing behavioral effects of post-training. Full results are in Appendix C.10.

4.3 Composability with other diversity interventions

Decoding- and prompt-based methods can increase output diversity without changing model parameters. We test whether SLR is complementary to two such inference-time interventions: higher-temperature min-p sampling ($T = 1.5$) [17] and verbalized sampling [38], which is a state-of-the-art prompt-based method that explicitly asks for K different responses and their corresponding probabilities in a single reply. For verbalized sampling, we use $K = 4$ candidates per call.

Results. Figure 4 summarizes the Open-ended QA results under higher-temperature decoding and verbalized sampling; full results, together with Creative Writing results under $T = 1.5$, are provided in Appendices C.2 and C.3. SLR increases entropy and coverage under higher-temperature decoding and VS4. Plain SLR yields higher precision, entropy, and coverage than Post-trained + VS4 for Llama and Gemma. For Qwen, Post-trained + VS4 yields higher entropy and coverage at the same reported precision. Combining SLR with VS4 further increases entropy and coverage across all three models, with lower precision than plain SLR. Together with the higher-temperature results, this provides an affirmative answer to Q4: *SLR is complementary to decoding- and prompt-based diversity interventions*.

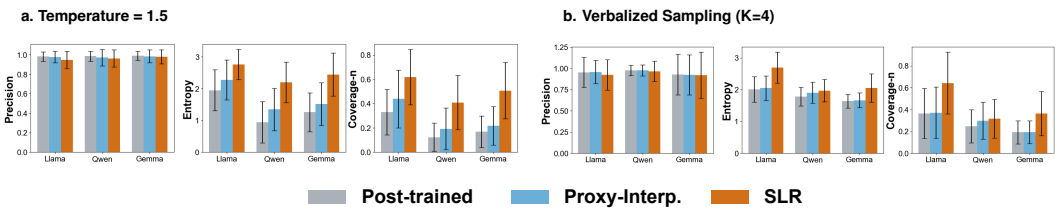


Figure 4: **Composability experiment results.** (a) higher sampling temperature ($T = 1.5$) and (b) verbalized sampling ($K = 4$). SLR continues to improve answer diversity, measured by entropy and Coverage-n, while preserving high precision.

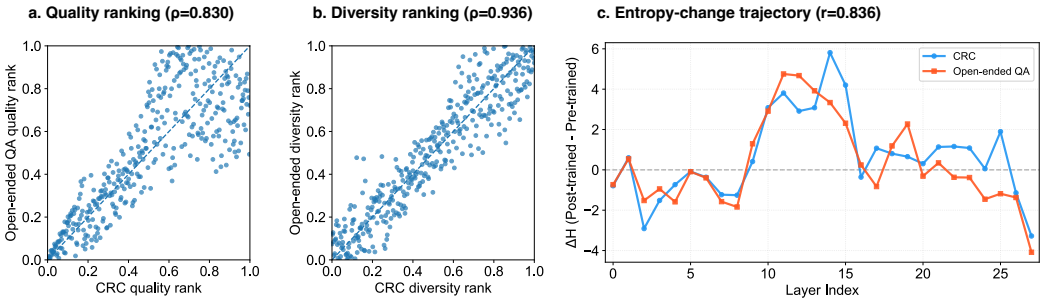


Figure 5: **CRC transfer analysis on Qwen.** (a–b) Rank-level agreement between CRC and Open-ended QA metrics across restoration intervals. (c) Logit-lens entropy-change trajectories, computed as $\Delta H_\ell = H_\ell(M_{\text{post}}) - H_\ell(M_{\text{pre}})$. Panel titles report the corresponding correlations.

4.4 Validating CRC as a transferable proxy

We test interval-ranking transfer from CRC to Open-ended QA, using Qwen as a representative case study. For creative writing and GSM8K, we evaluate the downstream performance of the CRC-selected intervals without task-specific interval tuning. Additional analyses, including results for Llama and Gemma, are provided in Appendix C.8.

Interval-level transfer. We first conduct a direct interval sweep on Open-ended QA and compare interval rankings under CRC and downstream metrics. For each restoration interval, we evaluate its CRC quality and diversity scores, as well as its downstream Open-ended QA scores to compare rank-level agreement between the proxy and downstream metrics. As shown in Figure 5(a), CRC quality rankings correlate with downstream precision rankings (Spearman $\rho = 0.830$). Figure 5(b) further shows that CRC diversity rankings strongly correlate with downstream diversity rankings (Spearman $\rho = 0.936$). The same analysis on Llama and Gemma also shows strong rank agreement (Appendix C.8), supporting CRC-guided interval selection for Open-ended QA across all three model families.

Logit-lens probe. We further apply a logit-lens probe [18] to provide more insights into this transfer. For each layer, we apply the model's final normalization and unembedding head to the hidden state at the final prompt token, and compute the entropy of the resulting next-token distribution. We measure $\Delta H_\ell = H_\ell(M_{\text{post}}) - H_\ell(M_{\text{pre}})$ to isolate the entropy change induced by post-training at each depth. As shown in Figure 5(c), CRC and Open-ended QA exhibit similar entropy-change trajectories across model depth on Qwen (with Pearson correlation $r = 0.836$). This suggests that CRC captures a depth-dependent distributional effect that also appears in downstream generation.

Comparison with pre-trained distributions. We also compare the pre-trained, post-trained, and SLR models under identical digit-completion prompts. For Llama and Qwen, SLR moves the distribution over valid digits closer to the pre-trained model under both Jensen–Shannon divergence and total variation distance while retaining high valid-token probability (Appendix C.13).

5 Conclusion

In this work, we presented Selective Layer Restoration (SLR), a training-free weight-space intervention for mitigating mode collapse in post-trained LLMs by restoring a contiguous interval of layers to their pre-trained parameters. We also introduced Constrained Random Character (CRC), a proxy task with an explicit validity set and a natural diversity objective that makes interval selection tractable and transferable.

SLR increases diversity in creative writing and open-ended QA with little loss in writing quality or answer accuracy. It also improves GSM8K Pass@k and composes with decoding- and prompt-based diversity interventions. Results on Open-ended QA with larger models (Appendix C.6) also demonstrate SLR's scalability. Our CRC transfer analysis further shows that CRC rankings align with Open-ended QA rankings, and that the two tasks exhibit similar logit-lens entropy-change trajectories across depth.

At a higher level, the effectiveness of SLR echoes and reinforces two insights: (i) it supports a modular view of LLMs, that their functionalities are not distributed evenly across layers or components [16, 25], and (ii) much of the LLMs' knowledge is acquired during pre-training and post-training primarily brings certain modes of behavior forward [11, 13, 29]. SLR suggests that fine-grained weight-space interventions offer a promising path toward both improved control and a deeper understanding of the quality–diversity tradeoff induced by post-training.

Limitations and future work. Our main SLR configuration restores entire transformer blocks. Appendix C.9 provides a preliminary module-level ablation, suggesting that MLP restoration accounts for much of the high-quality CRC diversity gain. Future work could explore layer-wise interpolation and finer-grained restoration of individual attention heads or FFN components. This paper restricts restoration candidates to contiguous layer intervals for tractability. Sparse, non-contiguous restoration may yield a better diversity–quality frontier. Efficient search over such subsets remains future work. SLR requires access to both the pre-trained and post-trained checkpoints. Restoring pre-trained layers can weaken instruction following and alter refusal behavior, with risks of over-refusal or compliance with unsafe requests (Appendix C.10). CRC-guided interval selection could account for sequence-level compliance and these behaviors alongside diversity. Our downstream evaluation is limited to English-language open-ended generation and reasoning benchmarks. Creative-writing quality scores and open-ended QA canonicalization rely on Gemini-2.5-Pro and are subject to its judgment errors and biases.

Future work could combine objectives that preserve multiple high-quality responses with more diverse post-training data. Monitoring layer-wise entropy during training could help locate diversity loss as it develops and guide where to constrain updates.

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A First-order view of SLR

Here, we provide a local and first-order analysis of SLR.

A residual transformer block maps a hidden state h_ℓ to

$$h_{\ell+1} = h_\ell + F_\ell(h_\ell).$$

Post-training changes the block map from F_ℓ^{pre} to F_ℓ^{post}. Restoring layer ℓ therefore changes the next hidden state by

$$\delta h_{\ell+1} = F_\ell^{\mathrm{pre}}(h_\ell) - F_\ell^{\mathrm{post}}(h_\ell).$$

For a fixed prefix, let H denote the entropy of the final next-token distribution. Let $\delta H_\ell^{\mathrm{restore}}$ denote the entropy after restoring layer ℓ minus that of the post-trained model. A local first-order approximation gives

$$\delta H_\ell^{\mathrm{restore}} \approx \left\langle \nabla_{h_{\ell+1}} H,\, F_\ell^{\mathrm{pre}}(h_\ell) - F_\ell^{\mathrm{post}}(h_\ell) \right\rangle.$$

Thus, SLR can increase output entropy when the restored layer removes a post-training update that locally sharpens the next-token distribution. Since both the update and the entropy gradient vary across depth, diversity loss need not be uniform across layers.

At the response level, the KL-regularized RLHF objective is equivalent to minimizing $D_{\mathrm{KL}}(\pi_\theta \| q)$, where $q(y \mid x) \propto \pi_{\mathrm{ref}}(y \mid x) \exp(r(x,y)/\beta)$, r is the reward, and $\beta > 0$ is the regularization coefficient [33]. The exponential weighting amplifies reward differences relative to β and can concentrate probability on a few responses. Reverse-KL approximation can also be mode-seeking when the policy family cannot represent the full target. Thus, mode collapse depends on both the target distribution and its approximation [9]. How these training effects become localized to particular layers remains an open question.

B Experimental details

B.1 Experiment settings

Generation hyperparameters. To ensure fairness and reproducibility, we fix the decoding parameters in our main experiments with a temperature of 1.0 and min-p sampling with $p_{base} = 0.1$. For creative writing and reasoning, the maximum output length is 4096 new tokens, while for open-ended QA, we limit to 64 tokens as the valid answers are all very short. In the temperature composability experiment, we only change the temperature to 1.5, leaving everything else unchanged. In the prompting composability experiment on open-ended QA, the temperature is set back to 1.0, and $K = 4$ responses are requested per call. Since the number of requested answers has increased, we also raised the maximum output length to 1024.

GSM8K evaluation. We select each model family's greedy-failure subset using only the post-trained model's greedy responses. All methods within a model family are evaluated on the same questions. We take the reference answer from the value following `####`. For each sampled response, we extract the last signed numerical value, remove formatting characters such as commas, currency symbols, and terminal punctuation, and compare it with the normalized reference by exact match. Responses without an extractable numerical answer are counted as incorrect. Given c correct answers among $n = 64$ samples for a question, we estimate Pass@k as $1 - \binom{n-c}{k} / \binom{n}{k}$ and average over questions.

Compute resources. The experiments in this paper were mainly conducted on a local server with an AMD EPYC 7763 64-core CPU and a single NVIDIA A100 GPU with 80GB memory. SLR is training-free, so the main computational cost comes from local generation for evaluation. On this machine, a complete run of the three main evaluation suites takes approximately 20 hours for Creative Writing, 1.5 hours for Open-ended QA, and 40 hours for Reasoning. The runtime is roughly the same across Post-trained, Proxy-Interpolation, and SLR, since all methods use the same model architecture, parameter count, and decoding settings; external API latency for LLM judging or canonicalization is not included in these estimates.

B.2 LLM-as-a-judge settings

We follow Zhang et al. [38] exactly for our LLM-as-a-judge setup, except we use Gemini-2.5-Pro as the judge model instead of Claude 3.7 Sonnet. The detailed rubric is:

Poem and Story Rubric

You are an expert in assessing creative writing. Your task is to score the quality of the test model's response below, by several metrics, on a 0-20 scale.
[PROMPT START]
{query}
[PROMPT END]
[TEST MODEL RESPONSE]
{response}
[TEST MODEL RESPONSE END]
[Task]
Scoring notes:
- A score of 20 represents a masterpiece.
- If the criteria is not relevant to the piece (e.g. criteria relating to dialogue but the piece has none), exclude it from your assessment.
- Do not add any commentary or explanation to the scores section. - Do not penalize NSFW output; that is the default for some models. Just assess it on its merits.
- Everything within the "TEST MODEL RESPONSE" section was written by the test model. Sometimes models like to write comments on the piece after the piece is concluded; if this happens you should ignore their comments.
- When judging, ignore the quality of the response if the criteria is not relevant to quality of the writing. - In the output, write the metric names exactly as below so they can be parsed.
- Do not use markdown in your response. Use the designated output format exactly. - You are to write a comprehensive analysis of the piece, then give your scores.
- You are a critic, and your job is to be critical, especially of any failings or amateurish elements.
- Output format is:
[Analysis]
Write your detailed analysis.
[Scores]
Metric 1 name: [Score 0-20]
Metric 2 name: ...
–
Now, rate the supplied model output on the following criteria:
1. Surprising and Creative 2. Imagery and Descriptive Quality 3. Nuanced Characters 4. Emotionally Complex 5. Elegant Prose 6. Well-earned Lightness or Darkness 7. Emotionally Engaging 8. Consistent Voice/Tone of Writing 9. Sentences Flow Naturally 10. Overall Reader Engagement

Joke

You will receive: 1. The original joke prompt (may or may not contain a topic). 2. The model-generated joke.
Your task is to evaluate the joke based on three qualitative metrics.
Evaluation rules:
- If the prompt includes a topic (e.g., "octopus," "coffee"), check whether the joke is on-topic and score Relevance from 0-5.
- If the prompt does not include a topic (e.g., "Tell me a joke"), automatically assign Relevance = 5.
- A good joke should use at least one recognizable comedic device (pun, irony, exaggeration, reversal, absurd logic, etc.).
- Assign scores on a 0-5 scale (0 = very poor, 5 = excellent) for each dimension:
- Relevance (0-5): How well does the joke address the topic (or 5 if no topic given).
- Comedic Device (0-5): How clearly does the joke use a humor mechanism.
- Humor Quality (0-5): How funny, witty, or clever is the joke overall.
Output format:
Return a JSON object in the following format:
"Relevance": ⟨ int ⟩,
"Comedic Device": ⟨ int ⟩,
"Humor Quality": ⟨ int ⟩
Input format:
Prompt: {query}
Generated joke: {response}

C Additional experimental results

C.1 Complete main experimental results

Table 1 presents the complete main experiment results on creative writing; Table 2 presents the complete main experiment results on open-ended QA; Table 3 presents the complete main experiment results on the subset of GSM8K where the corresponding model's greedy generation fails.

Table 1: Complete results of Post-trained model, SLR, and Proxy-Interp. on the Creative Writing task, including three subtasks, Joke, Poem, and Story generation. Quality is measured using LLM-as-a-judge; diversity is measured with embedding dissimilarity.

		Llama		**Qwen**		**Gemma**	
Dataset	Method	Quality (↑)	Diversity (↑)	Quality (↑)	Diversity (↑)	Quality (↑)	Diversity (↑)
Joke	Post-trained	12.6 (1.870)	0.251 (0.124)	11.6 (2.134)	0.095 (0.055)	12.0 (2.326)	0.208 (0.123)
	Proxy-Interp.	12.1 (1.946)	0.299 (0.125)	11.3 (2.130)	0.220 (0.053)	11.6 (2.209)	0.190 (0.048)
	SLR	12.8 (2.157)	0.332 (0.128)	11.6 (2.597)	0.190 (0.048)	11.5 (2.670)	0.294 (0.084)
Poem	Post-trained	79.0 (19.21)	0.144 (0.038)	88.2 (23.01)	0.152 (0.035)	104.3 (23.18)	0.127 (0.041)
	Proxy-Interp.	75.9 (19.15)	0.153 (0.040)	70.5 (24.44)	0.179 (0.033)	99.8 (23.93)	0.143 (0.045)
	SLR	75.4 (20.87)	0.240 (0.054)	87.6 (20.89)	0.190 (0.038)	100.5 (21.85)	0.171 (0.051)
Story	Post-trained	93.4 (20.32)	0.274 (0.082)	94.7 (21.51)	0.282 (0.072)	122.6 (22.41)	0.294 (0.092)
	Proxy-Interp.	87.8 (20.72)	0.297 (0.084)	80.4 (22.38)	0.297 (0.075)	117.5 (22.89)	0.326 (0.094)
	SLR	90.8 (23.58)	0.366 (0.079)	93.8 (21.15)	0.328 (0.069)	117.7 (27.97)	0.364 (0.070)

Table 2: Complete results of Post-trained, SLR, and Proxy-Interp. on the Open-ended QA task (broken down into 3 subtables for each model due to space constraints). Quality is measured with precision; diversity is measured with entropy over correct answers and Coverage-n (fraction of correct answers generated at least once). Higher is better for all metrics.

Llama			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained	0.986 (0.053)	1.442 (0.675)	0.204 (0.140)
Proxy-Interp.	0.981 (0.060)	1.760 (0.716)	0.285 (0.182)
SLR	0.968 (0.074)	2.322 (0.686)	0.461 (0.256)

Qwen			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained	0.999 (0.002)	0.659 (0.538)	0.086 (0.080)
Proxy-Interp.	0.977 (0.087)	0.919 (0.637)	0.124 (0.116)
SLR	0.978 (0.079)	1.583 (0.644)	0.226 (0.162)

Gemma			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained	0.994 (0.0264)	0.771 (0.595)	0.096 (0.081)
Proxy-Interp.	0.993 (0.0315)	0.967 (0.121)	0.121 (0.094)
SLR	0.978 (0.0961)	1.827 (0.751)	0.307 (0.209)

Table 3: Complete results of Post-trained, SLR, and Proxy-Interp. on the GSM8K subset. Pass@k with $k \in \{1, 4, 8, 16, 32, 64\}$ are reported to measure sampling success at different budgets. Higher is better. Best results are bolded.

	Llama			**Qwen**			**Gemma**		
	Post-trained	Proxy-Interp.	SLR	Post-trained	Proxy-Interp.	SLR	Post-trained	Proxy-Interp.	SLR
Pass@1	0.05	0.14	**0.15**	0.07	0.28	**0.32**	0.01	0.08	**0.21**
Pass@4	0.16	0.38	**0.43**	0.25	0.53	**0.60**	0.04	0.21	**0.46**
Pass@8	0.28	0.53	**0.61**	0.40	0.62	**0.72**	0.08	0.30	**0.59**
Pass@16	0.44	0.66	**0.78**	0.56	0.69	**0.79**	0.13	0.38	**0.71**
Pass@32	0.58	0.76	**0.89**	0.67	0.74	**0.85**	0.20	0.42	**0.79**
Pass@64	0.66	0.80	**0.95**	0.73	0.77	**0.91**	0.26	0.44	**0.84**

C.2 Additional temperature results

Table 4 presents results on creative writing under temperature $T = 1.5$; Table 5 presents results on Open-ended QA under temperature $T = 1.5$. Table 6 reports Open-ended QA results at $T = 0.5$.

Table 4: Complete results of Post-trained model, SLR, and Proxy-Interp. on the Creative Writing task, including three subtasks, Joke, Poem, and Story generation **at temperature** $T = 1.5$. Quality is measured using LLM-as-a-judge; diversity is measured with embedding dissimilarity.

		Llama		**Qwen**		**Gemma**	
Dataset	Method	Quality (↑)	Diversity (↑)	Quality (↑)	Diversity (↑)	Quality (↑)	Diversity (↑)
Joke	Post-trained	12.6(1.854)	0.321(0.122)	11.5 (2.125)	0.123 (0.058)	11.9 (2.370)	0.278 (0.117)
	Proxy-Interp.	12.0(1.979)	0.381(0.125)	11.0 (2.479)	0.282 (0.048)	11.4 (2.273)	0.327 (0.114)
	SLR	12.8(2.209)	0.406(0.128)	11.6 (2.782)	0.289 (0.054)	11.3 (2.941)	0.331 (0.069)
Poem	Post-trained	82.0 (20.04)	0.170 (0.043)	89.6 (24.14)	0.183 (0.036)	107.4 (24.04)	0.183 (0.040)
	Proxy-Interp.	77.9 (20.90)	0.184 (0.047)	72.1 (24.38)	0.220 (0.037)	103.8 (24.24)	0.167 (0.048)
	SLR	81.5 (20.23)	0.280 (0.053)	89.4 (23.65)	0.233 (0.041)	104.8 (23.43)	0.244 (0.057)
Story	Post-trained	102.9 (23.87)	0.314 (0.081)	103.5 (23.61)	0.324 (0.075)	130.0 (21.57)	0.317 (0.091)
	Proxy-Interp.	95.0 (23.56)	0.337 (0.085)	87.4 (21.34)	0.336 (0.076)	125.8 (22.12)	0.353 (0.094)
	SLR	102.6 (22.82)	0.406 (0.076)	103.2 (25.12)	0.385 (0.073)	135.6 (29.26)	0.405 (0.070)

Table 5: Complete results of Post-trained, SLR, and Proxy-Interp. on open-ended QA at **temperature** $T = 1.5$ (broken down into 3 subtables for each model due to space constraints). Quality is measured with precision; diversity is measured with entropy over correct answers and Coverage-n (fraction of correct answers generated at least once). Higher is better for all metrics.

Llama

Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained @$T = 1.5$	0.982 (0.049)	1.948 (0.643)	0.329 (0.188)
Proxy-Interp. @$T = 1.5$	0.977 (0.059)	2.275 (0.631)	0.438 (0.238)
SLR @$T = 1.5$	0.947 (0.090)	2.759 (0.476)	0.617 (0.228)

Qwen

Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained @$T = 1.5$	0.985 (0.052)	0.944 (0.652)	0.123 (0.116)
Proxy-Interp. @$T = 1.5$	0.971 (0.084)	1.345 (0.669)	0.192 (0.171)
SLR @$T = 1.5$	0.962 (0.088)	2.196 (0.641)	0.408 (0.224)

Gemma

Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained @$T = 1.5$	0.990 (0.046)	1.259 (0.612)	0.167 (0.130)
Proxy-Interp. @$T = 1.5$	0.984 (0.066)	1.515 (0.673)	0.217 (0.158)
SLR @$T = 1.5$	0.978 (0.071)	2.445 (0.681)	0.506 (0.231)

Table 6: Open-ended QA at temperature $T = 0.5$. Values are mean precision, entropy over correct answers, and Coverage-n. Best results within each model family are bolded.

Model	Method	Precision	Entropy	Coverage-n
Llama	Post-trained	**0.988**	0.811	0.099
	Proxy-Interp.	0.980	1.101	0.136
	SLR	0.981	**1.416**	**0.207**
Qwen	Post-trained	**1.000**	0.275	0.044
	Proxy-Interp.	0.982	0.392	0.058
	SLR	0.976	**0.726**	**0.088**
Gemma	Post-trained	0.995	0.343	0.046
	Proxy-Interp.	**0.997**	0.460	0.059
	SLR	0.979	**0.866**	**0.113**

Model-specific temperatures. We evaluate Open-ended QA at $T = 0.6$ for Llama and $T = 0.7$ for Qwen, retaining $T = 1.0$ for Gemma. Relative to the post-trained model, SLR increases entropy and Coverage-n for all three models ($p_{\text{Holm}} < 0.001$), with no significant difference in precision ($p_{\text{Holm}} > 0.05$). Table 7 reports the results.

Table 7: Open-ended QA at model-specific temperatures. Values are mean precision, entropy over correct answers, and Coverage-n. Best results within each model family are bolded.

		Post-trained			**SLR**		
Model	T	Precision	Entropy	Coverage-n	Precision	Entropy	Coverage-n
Llama	0.6	**0.990**	0.958	0.121	0.984	**1.669**	**0.265**
Qwen	0.7	**1.000**	0.412	0.058	0.975	**1.116**	**0.133**
Gemma	1.0	**0.994**	0.771	0.096	0.978	**1.827**	**0.307**

Comparison at comparable precision. We compare SLR at $T = 1$ with temperature-tuned post-trained decoding at $T = 1.80$, 1.65, and 1.75 for Llama, Qwen, and Gemma, respectively. Evaluation follows the QA protocol in Section 4.1, and SLR precision values are those in Table 2. We compute 95% CIs using 100,000 paired prompt bootstraps and two-sided p-values using 200,000 sign-flip draws over the same 40 prompts. Holm correction covers the six diversity tests. In Table 8, SLR has higher entropy and coverage for all three models; the gains remain significant after Holm correction for Qwen and Gemma, while Llama has $p_{\mathrm{Holm}} = 0.0562$ for both metrics.

Table 8: Open-ended QA at comparable precision. The upper panel reports precision for SLR ($T = 1$) and temperature-tuned Post-trained. The lower panel reports prompt-paired differences, SLR minus Post-trained. Confidence intervals are pointwise 95% intervals. p-values are two-sided; p_{Holm} adjusts for all six diversity comparisons.

Model	Post-trained T	SLR precision	Post-trained precision
Llama	1.80	0.968	0.975
Qwen	1.65	0.978	0.979
Gemma	1.75	0.978	0.973

Model	Metric	Δ	95% CI	p	p_{Holm}
Llama	Entropy	$+0.134$	$[+0.006, +0.255]$	0.0426	0.0562
Llama	Coverage-n	$+0.051$	$[+0.008, +0.095]$	0.0281	0.0562
Qwen	Entropy	$+0.582$	$[+0.426, +0.740]$	$\leq 5.00 \times 10^{-6}$	$\leq 3.00 \times 10^{-5}$
Qwen	Coverage-n	$+0.097$	$[+0.072, +0.122]$	$\leq 5.00 \times 10^{-6}$	$\leq 3.00 \times 10^{-5}$
Gemma	Entropy	$+0.265$	$[+0.104, +0.429]$	0.00285	0.00855
Gemma	Coverage-n	$+0.074$	$[+0.034, +0.117]$	0.000600	0.00240

C.3 Complete experimental results with Verbalized Sampling

Table 9 compares plain SLR with verbalized sampling ($K = 4$) applied to the post-trained, Proxy-Interpolation, and SLR models. For VS4, we make 32 calls per prompt and evaluate four answers from each call, yielding 128 answers per prompt, matching the plain-prompt setting.

Table 9: Open-ended QA comparison of plain SLR and verbalized sampling ($K = 4$). Plain SLR results are repeated from Table 2. Quality is measured with precision; diversity is measured with entropy over correct answers and Coverage-n (fraction of correct answers generated at least once). Higher is better for all metrics.

Llama			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
SLR (plain)	0.968 (0.074)	2.322 (0.686)	0.461 (0.256)
Post-trained + VS4	0.954 (0.176)	2.008 (0.399)	0.365 (0.228)
Proxy-Interp. + VS4	0.957 (0.137)	2.047 (0.389)	0.371 (0.235)
SLR + VS4	0.924 (0.180)	2.693 (0.487)	0.643 (0.283)

Qwen			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
SLR (plain)	0.978 (0.079)	1.583 (0.644)	0.226 (0.162)
Post-trained + VS4	0.978 (0.061)	1.779 (0.296)	0.248 (0.151)
Proxy-Interp. + VS4	0.975 (0.065)	1.905 (0.327)	0.299 (0.170)
SLR + VS4	0.965 (0.120)	1.970 (0.354)	0.317 (0.176)

Gemma			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
SLR (plain)	0.978 (0.0961)	1.827 (0.751)	0.307 (0.209)
Post-trained + VS4	0.926 (0.240)	1.642 (0.218)	0.192 (0.105)
Proxy-Interp. + VS4	0.922 (0.236)	1.663 (0.232)	0.194 (0.105)
SLR + VS4	0.919 (0.273)	2.055 (0.443)	0.363 (0.202)

C.4 Qualitative results

We also present some qualitative examples here to illustrate the effect of SLR: Tables 10–13 show example joke, poem, story, and open-ended QA results, respectively. A reasoning example where the post-trained model is unable to find the correct answer while SLR succeeds is also presented in Table 14.

Table 10: Example generations for a joke prompt, using Llama3.1-8B.

Prompt: "Tell me a joke."	
Post-trained	**SLR**
A man walked into a library and asked the librarian, "Do you have any books on Pavlov's dogs and Schrodinger's cat?" The librarian replied, "It rings a bell, but I'm not sure if it's here or not."	A man walked into a library and asked the librarian, "Do you have any books on Pavlov's dogs and Schrodinger's cat?" The librarian replied, "It rings a bell, but I'm not sure if it's here or not."
A man walked into a library and asked the librarian, "Do you have any books on Pavlov's dogs and Schrodinger's cat?" The librarian replied, "It rings a bell, but I'm not sure if it's here or not."	What do you call a fake noodle? An impasta!
A man walked into a library and asked the librarian, "Do you have any books on Pavlov's dogs and Schrodinger's cat?" The librarian replied, "It rings a bell, but I'm not sure if it's here or not."	Why don't scientists trust atoms? Because they make up everything!

Table 11: Example generations for a poem prompt, using Llama3.1-8B.

Prompt: "Write a poem."	
Post-trained	**SLR**
Whispers of the Past In twilight's hush, where shadows play, A whispered secret echoes through the day. A tale of love and loss, of trials and strife, A story woven from the threads of life...	In the depths of my soul, a spark ignites, A flame that flickers, yet never fades. A light that guides me, through darkest nights, A beacon in the darkness, that shines through shades...
Whispers of the Wind The wind whispers secrets in my ear, Of distant lands and memories so dear. It carries scents of blooms and trees, A symphony of nature's breeze. With gentle touch, it stirs the leaves, A soft caress that brings me to my knees...	In twilight's hush, where shadows play, A midnight breeze whispers through the day. The stars above, a celestial sea, Reflect the dreams that dance within me...
Whispers in the wind, a gentle breeze, Dances through the trees, with leaves that sway. The sun shines bright, a warm and golden light, Illuminating all, in a radiant display...	Whispers of Memories In the silence of the night, A gentle breeze stirs the trees, And I am lost in thought, As memories of you dance with the breeze...

Table 12: Example generations for a story prompt, using Llama3.1-8B.

Prompt: "Write a story."	
Post-trained	**SLR**
Once upon a time, in a small village nestled in the rolling hills of the countryside, there lived a young girl named Luna. She was a curious and adventurous soul, with a heart full of wonder and a mind full of questions...	**The Mysterious Stranger** The sun was setting over the small town of Willow Creek, casting a warm orange glow over the quaint streets and homes. The air was filled with the sweet scent of blooming flowers, and the sound of children's laughter echoed through the streets as they played in the park...
Once upon a time, in a small village nestled in the rolling hills of a lush countryside, there lived a young girl named Luna. Luna was a curious and adventurous soul, with a heart full of wonder and a mind full of questions...	The Island of Lost Things As the sun dipped below the horizon, casting a warm orange glow over the calm waters of the Mediterranean, Captain James Blackwood stood at the helm of his ship, gazing out at the endless expanse of sea...
In the small town of Willow Creek, nestled in the heart of the Whispering Woods, there lived a young girl named Luna. She was a curious and adventurous soul, with eyes as bright as the stars and hair as wild as the wind...	Once upon a time, in a small village nestled in the rolling hills of rural England, there lived a young girl named Emily. She was a bright and curious child, with a wild mane of curly brown hair and a mischievous twinkle in her eye...

Table 13: Example generations for an open-ended QA prompt, using Llama3.1-8B.

Prompt: "Name a National Park of the United States. Only provide the answer without explanation or punctuation."	
Post-trained	**SLR**
Yellowstone	Acadia National Park
Yellowstone National Park	Yosemite
Yellowstone	Everglades National Park
Yellowstone	Rocky Mountain National Park
Yellowstone National Park	Great Smoky Mountains National Park
Yellowstone National Park	Glacier National Park
Yellowstone National Park	Grand Teton National Park
Yellowstone	Yellowstone National Park

Table 14: An example reasoning question where the post-trained model is unable to find the correct answer in 64 samples while SLR succeeds, using Llama3.1-8B.

Prompt: "A company pays each of its employees $600 in a month. The company has a policy of increasing the salaries of each of its employees by 10% of the initial salary every year for those who've stayed in the company for five years. If Sylvie just clocked 5 years in the company last December, what's her annual salary after three more years of service?"

In a month, Sylvie earns $600, which totals $600/month $\times$ 12 months/year $= \$7200$ in a year. Sylvie's earnings will increase by $\frac{10}{100} \times \$7200/year = \$720/year$ after 5 years. In three years, the total increase in her salary will be 3 *years* $\times \$720/year = \$2160$. *After three years, Sylvie will be paid* $\$7200 + \$2160 = \$9360$.

Post-trained	**SLR**
To find Sylvie's annual salary after three more years of service, we need to calculate the salary after the 10% increase, which happens every year for 5 years. Her initial annual salary is $600. 1. Increase after 1st year: $600 + (10% of $600) = $600 + $60 = $660 2. Increase after 2nd year: $660 + (10% of $660) = $660 + $66 = $726 3. Increase after 3rd year: $726 + (10% of $726) = $726 + $72.6 = $798.6 4. Increase after 4th year: $798.6 + (10% of $798.6) = $798.6 + $79.86 = $878.46 5. Increase after 5th year: $878.46 + (10% of $878.46) = $878.46 + $87.846 = $966.306 After the 5th year, Sylvie's annual salary becomes $966.31. For the next three years, she will still be getting a 10% increase every year. 6. Increase after 6th year: $966.31 + (10% of $966.31) = $966.31 + $96.631 = $1062.941 7. Increase after 7th year: $1062.94 + (10% of $1062.94) = $1062.94 + $106.294 = $1169.234 8. Increase after 8th year: $1169.23 + (10% of $1169.23) = $1169.23 + $116.923 = $1286.153 After three more years of service, Sylvie's annual salary will be approximately $1286.15.	To calculate Sylvie's salary after 3 years, we first need to find out her initial salary. We are given that the company pays each employee $600 in a month. Assuming there are 12 months in a year, the annual salary of Sylvie would be 600 * 12 = $7,200. Now, since Sylvie stayed in the company for 5 years, her salary would increase by 10% of the initial salary every year for the next three years. 10% of $7,200 = (10/100) * 7,200 = $720. So, after the first year, Sylvie's salary would be $7,200 + $720 = $7,920. Now, to calculate her salary for the next two years, we will add $720 to the previous year's salary. After 2 years, her salary would be $7,920 + $720 = $8,640. After 3 years, her salary would be $8,640 + $720 = $9,360.

C.5 Additional weight-space baselines

Table 15 compares SLR with additional weight-space baselines on Open-ended QA. Vanilla-Soup uses a fixed global interpolation coefficient $\alpha = 0.5$, which moves every layer halfway from the post-trained checkpoint toward the pre-trained checkpoint. For Qwen, CRC selects $\alpha = 0.5$, so Proxy-Interpolation coincides with Vanilla-Soup. Fixed interpolation incurs large precision drops for Llama and Gemma, with Llama falling from 0.986 to 0.311. With a mixing budget of 0.2, HMA yields precision of 0.165, 0.704, and 0.814 for Llama, Qwen, and Gemma, respectively. Proxy-Interpolation and SLR are both selected using CRC. SLR restores the interval that maximizes CRC diversity under the quality constraint and yields higher entropy and coverage than Proxy-Interpolation, with higher precision than HMA across all three model families.

Table 15: Complete results of Post-trained, SLR, Proxy-Interp., HMA (mixing budget=0.2), and Vanilla-Soup ($\alpha = 0.5$) on the Open-ended QA task. Quality is measured with precision; diversity is measured with entropy over correct answers and Coverage-n (fraction of correct answers generated at least once). Higher is better for all metrics.

Llama			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained	0.986	1.442	0.204
Proxy-Interp.	0.981	1.760	0.285
SLR	0.968	2.322	0.461
HMA	0.165	1.336	0.140
Vanilla-Soup	0.311	1.860	0.296

Qwen			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained	0.999	0.659	0.086
Proxy-Interp.	0.977	0.919	0.124
SLR	0.978	1.583	0.226
HMA	0.704	1.260	0.161
Vanilla-Soup	0.977	0.919	0.124

Gemma			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained	0.994	0.771	0.096
Proxy-Interp.	0.993	0.967	0.121
SLR	0.978	1.827	0.307
HMA	0.814	1.856	0.281
Vanilla-Soup	0.791	1.408	0.201

C.6 Larger-model open-ended QA results

For the larger-model experiments, CRC selects restoration interval [22, 44] for Gemma2-27B and [28, 59] for Qwen2.5-32B, using the same criteria as stated in Section 3.3. Table 16 shows that the same qualitative pattern persists on larger models: the post-trained models remain highly precise but low-diversity, while SLR substantially improves entropy and Coverage-n with only a modest precision drop. SLR also improves diversity over Proxy-Interpolation on both Gemma2-27B and Qwen2.5-32B, suggesting that targeted restoration remains more effective than global interpolation at a larger scale. Overall, these results provide evidence that SLR scales beyond the 7–9B regime, though a broader scaling study across more tasks and model sizes remains future work.

Table 16: Complete larger-model results of Post-trained, SLR, and Proxy-Interp. on the Open-ended QA task. Quality is measured with precision; diversity is measured with entropy over correct answers and Coverage-n (fraction of correct answers generated at least once). Higher is better for all metrics.

Gemma2-27B			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained	0.998	0.658	0.081
Proxy-Interp.	0.976	0.866	0.109
SLR	0.968	1.605	0.236

Qwen2.5-32B			
Method	Precision(↑)	Entropy(↑)	Coverage-n(↑)
Post-trained	0.999	0.317	0.044
Proxy-Interp.	0.932	0.675	0.076
SLR	0.964	0.997	0.121

C.7 Random GSM8K subset results

We evaluate all three model families on the same random subset of 100 GSM8K questions, sampled without conditioning on greedy-decoding outcomes. We did not redraw the subset or repeat the

evaluation with different generation seeds. SLR improves Pass@k over both baselines at every sampling budget (Table 17), showing that its gains persist beyond the greedy-failure setting.

Table 17: GSM8K Pass@k on the same random subset of 100 questions for Llama, Qwen, and Gemma. Higher is better. Best results within each model family are bolded.

	Llama			**Qwen**			**Gemma**		
Metric	Post-trained	Proxy-Interp.	SLR	Post-trained	Proxy-Interp.	SLR	Post-trained	Proxy-Interp.	SLR
Pass@1	0.16	0.22	**0.23**	0.18	0.34	**0.36**	0.13	0.17	**0.28**
Pass@4	0.38	0.51	**0.55**	0.44	0.65	**0.68**	0.27	0.38	**0.57**
Pass@8	0.50	0.64	**0.70**	0.58	0.75	**0.78**	0.33	0.46	**0.69**
Pass@16	0.63	0.74	**0.83**	0.70	0.80	**0.84**	0.39	0.53	**0.78**
Pass@32	0.73	0.82	**0.92**	0.78	0.83	**0.89**	0.46	0.56	**0.84**
Pass@64	0.78	0.85	**0.96**	0.81	0.86	**0.93**	0.53	0.58	**0.88**

C.8 Additional CRC analysis

Figure 6 extends the interval-ranking analysis to Llama and Gemma. CRC quality correlates with Open-ended QA precision for Llama ($\rho = 0.802$) and Gemma ($\rho = 0.750$), while CRC diversity correlates with answer entropy at $\rho = 0.802$ and $\rho = 0.795$, respectively.

Figures 7 and 8 extend the logit-lens entropy-change analysis to Llama and Gemma. For each layer, we compute $\Delta H = H(M_{\text{post}}) - H(M_{\text{pre}})$ for both CRC and Open-ended QA prompts. The trajectories are positively correlated for both Llama ($r = 0.766$, $p = 3.31 \times 10^{-7}$) and Gemma ($r = 0.702$, $p = 2.22 \times 10^{-7}$), suggesting that CRC captures a depth-dependent distributional effect that also appears in downstream Open-ended QA. Together with the Qwen interval-ranking analysis in the main text, these results support the use of CRC as a low-cost structural proxy.

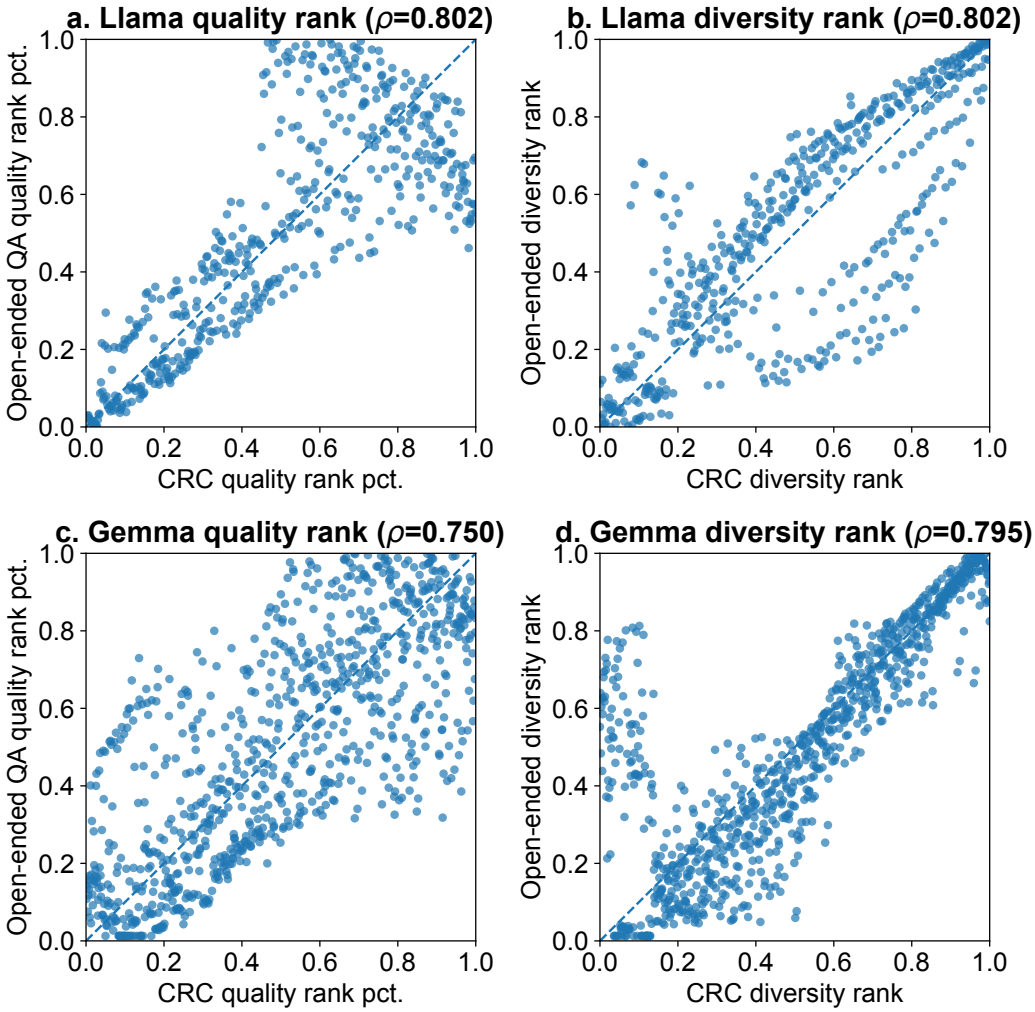


Figure 6: CRC–Open-ended QA rank agreement on Llama and Gemma. Each point represents a restoration interval. Axes show percentile ranks, with average ranks for ties. Dashed diagonals indicate identical ranks.

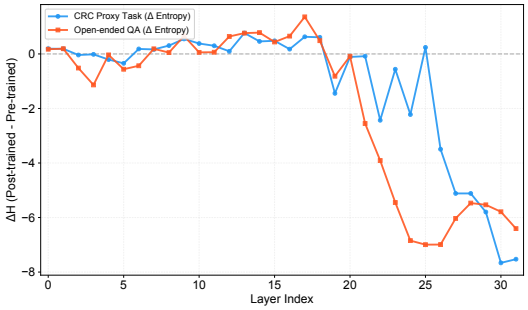


Figure 7: Llama-3.1-8B entropy-change trajectory. CRC and Open-ended QA show similar depth-dependent entropy changes, with a Pearson correlation of $r = 0.766$ ($p = 3.31 \times 10^{-7}$).

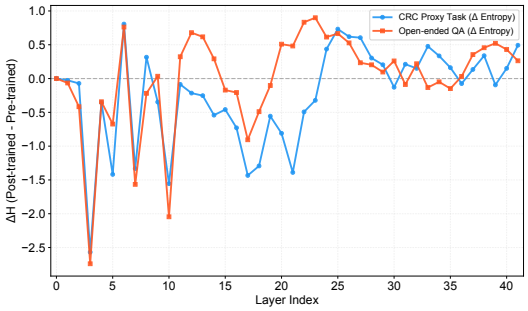


Figure 8: Gemma2-9B entropy-change trajectory. CRC and Open-ended QA show similar depth-dependent entropy changes, with a Pearson correlation of $r = 0.702$ ($p = 2.22 \times 10^{-7}$).

C.9 MLP/attention ablation study

Figures 9, 10, and 11 compare CRC Pareto frontiers from restoring full transformer blocks, only MLP modules, or only attention modules. Across all three model families, attention-only restoration gives a weaker diversity–quality frontier, while MLP-only restoration is much closer to full-block restoration. This suggests that much of SLR's high-quality diversity gain is mediated through the MLP pathway, although full-block restoration can still be preferable when optimizing the overall frontier.

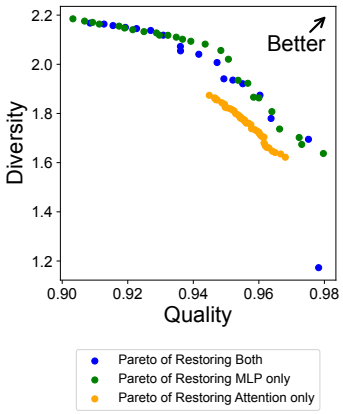


Figure 9: Llama-3.1-8B module ablation on CRC. MLP-only restoration closely matches full-block restoration, while attention-only restoration is weaker.

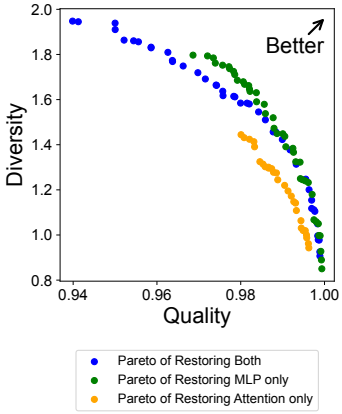


Figure 10: Qwen2.5-7B module ablation on CRC. MLP-only restoration is much stronger than attention-only restoration and approaches full-block restoration.

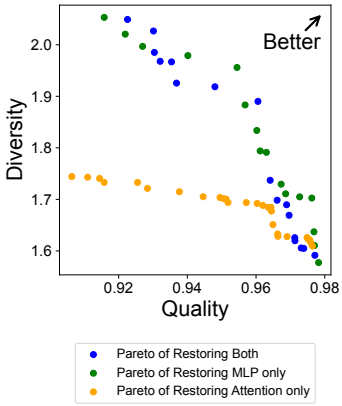


Figure 11: Gemma2-9B module ablation on CRC. MLP-only restoration is close to full-block restoration, while attention-only restoration gives smaller diversity gains.

C.10 Instruction-following and refusal diagnostics

Tables 18 and 19 report instruction-following accuracy and refusal behavior under greedy decoding. On IFEval, SLR lowers instruction-following accuracy relative to the post-trained model across all three model families. Compared with Proxy-Interpolation, SLR is stronger on Llama but weaker on Qwen and Gemma. On XSTest, the effect is model-dependent: SLR slightly worsens refusal behavior for Llama, increases unsafe-prompt refusal for Qwen at the cost of more over-refusal on safe prompts, and reduces over-refusal for Gemma while also reducing unsafe-prompt refusal. These results suggest that SLR can shift alignment and refusal behavior in non-uniform ways, so deployment of SLR-restored checkpoints would require task- and model-specific safety evaluation.

C.11 CRC threshold sensitivity

We vary the multiplier τ in $q_{\min} = \tau \max_{M \in \mathcal{C}} Q_{\mathrm{CRC}}(M)$ while retaining the same interval-selection rule. Table 20 reports the selected intervals and their open-ended QA results.

Table 18: IFEval results for Post-trained, SLR, and Proxy-Interp. Higher is better for all metrics.

Model	Method	Strict Prompt ↑	Strict Instr. ↑	Loose Prompt ↑	Loose Instr. ↑
Llama-3.1-8B	Post-trained	0.752	0.824	0.793	0.855
	SLR	0.627	0.723	0.695	0.777
	Proxy-Interp.	0.543	0.663	0.610	0.716
Qwen2.5-7B	Post-trained	0.725	0.800	0.743	0.818
	SLR	0.532	0.639	0.567	0.669
	Proxy-Interp.	0.601	0.697	0.662	0.747
Gemma2-9B	Post-trained	0.739	0.811	0.758	0.827
	SLR	0.538	0.650	0.573	0.675
	Proxy-Interp.	0.669	0.764	0.702	0.789

Table 19: XSTest refusal rates for Post-trained, SLR, and Proxy-Interp. Safe prompts contain 250 examples, where lower refusal rate is better; unsafe prompts contain 200 examples, where higher refusal rate is better. Best results are bolded.

	Llama		**Qwen**		**Gemma**	
Method	Safe Refusal ↓	Unsafe Refusal ↑	Safe Refusal ↓	Unsafe Refusal ↑	Safe Refusal ↓	Unsafe Refusal ↑
Post-trained	**6.0%**	**94.5%**	**4.0%**	66.0%	8.4%	73.5%
Proxy-Interp.	**6.0%**	93.0%	9.6%	82.5%	12.8%	**77.5%**
SLR	8.4%	91.5%	13.6%	**89.5%**	**6.8%**	67.0%

Table 20: CRC threshold sensitivity. Restored fractions are percentages of transformer layers. Layer indices are zero-based and inclusive.

Model	τ	Interval	Restored (%)	Precision	Entropy	Coverage-n
Llama	0.900	$[12, 28]$	53.1	0.943	2.605	0.550
	0.925	$[12, 23]$	37.5	0.956	2.455	0.499
	0.950	$[12, 17]$	18.8	0.968	2.322	0.461
	0.975	$[15, 18]$	12.5	0.982	1.736	0.270
	0.990	$[4, 6]$	9.4	0.982	1.375	0.196
Qwen	0.900	$[8, 27]$	71.4	0.886	1.679	0.245
	0.925	$[9, 27]$	67.9	0.909	1.656	0.244
	0.950	$[12, 27]$	57.1	0.978	1.583	0.226
	0.975	$[12, 22]$	39.3	0.980	1.261	0.169
	0.990	$[10, 19]$	35.7	0.986	1.063	0.136
Gemma	0.900	$[23, 41]$	45.2	0.962	2.452	0.489
	0.925	$[19, 30]$	28.6	0.975	2.187	0.395
	0.950	$[17, 27]$	26.2	0.978	1.827	0.307
	0.975	$[18, 25]$	19.0	0.982	1.541	0.228
	0.990	$[17, 22]$	14.3	0.989	1.154	0.148

C.12 Statistical significance of the main results

Tables 21 and 22 give the detailed creative-writing and open-ended QA statistical results at $T = 1$. Effects are SLR minus the specified baseline, with Post-trained as the primary comparison and Proxy-Interpolation as the secondary comparison. We report raw two-sided p-values and Holm-adjusted values, assessing significance with $p_{\text{Holm}} < 0.05$. For each baseline comparison, the correction families contain nine creative-writing quality tests, nine creative-writing diversity tests, three QA precision tests, and six QA diversity tests. Confidence intervals are pointwise 95% intervals without multiplicity correction.

Creative writing and open-ended QA. Tables 21 and 22 use the prompt-level means and standard deviations in Tables 1 and 2, with 100 prompts per creative-writing task and 40 QA prompts. The confidence intervals and raw p-values use Welch inference from these summary statistics.

Table 21: Creative-writing differences between SLR and each baseline at $T = 1$. Each metric reports Δ [pointwise 95% CI], the raw two-sided Welch p-value, and its Holm-adjusted value. Quality differences use the original judge-score units; diversity differences use embedding dissimilarity.

Task	Model	Δ [95% CI]	p_{raw}	p_{Holm}
Quality: SLR − Post-trained				
Joke	Llama	$+0.2000$ $[-0.3630, +0.7630]$	0.4844	1.0000
Joke	Qwen	$+0.0000$ $[-0.6630, +0.6630]$	1.0000	1.0000
Joke	Gemma	-0.5000 $[-1.1984, +0.1984]$	0.1595	1.0000
Poem	Llama	-3.6000 $[-9.1939, +1.9939]$	0.2059	1.0000
Poem	Qwen	-0.6000 $[-6.7290, +5.5290]$	0.8471	1.0000
Poem	Gemma	-3.8000 $[-10.0820, +2.4820]$	0.2343	1.0000
Story	Llama	-2.6000 $[-8.7392, +3.5392]$	0.4046	1.0000
Story	Qwen	-0.9000 $[-6.8488, +5.0488]$	0.7658	1.0000
Story	Gemma	-4.9000 $[-11.9698, +2.1698]$	0.1732	1.0000
Quality: SLR − Proxy-Interp.				
Joke	Llama	$+0.7000$ $[+0.1271, +1.2729]$	0.0169	0.1183
Joke	Qwen	$+0.3000$ $[-0.3625, +0.9625]$	0.3729	1.0000
Joke	Gemma	-0.1000 $[-0.7835, +0.5835]$	0.7732	1.0000
Poem	Llama	-0.5000 $[-6.0859, +5.0859]$	0.8601	1.0000
Poem	Qwen	$+17.1000$ $[+10.7588, +23.4412]$	2.87×10^{-7}	2.58×10^{-6}
Poem	Gemma	$+0.7000$ $[-5.6906, +7.0906]$	0.8292	1.0000
Story	Llama	$+3.0000$ $[-3.1908, +9.1908]$	0.3404	1.0000
Story	Qwen	$+13.4000$ $[+7.3275, +19.4725]$	2.17×10^{-5}	1.74×10^{-4}
Story	Gemma	$+0.2000$ $[-6.9291, +7.3291]$	0.9559	1.0000
Diversity: SLR − Post-trained				
Joke	Llama	$+0.0810$ $[+0.0459, +0.1161]$	9.55×10^{-6}	1.43×10^{-5}
Joke	Qwen	$+0.0950$ $[+0.0806, +0.1094]$	2.95×10^{-28}	2.36×10^{-27}
Joke	Gemma	$+0.0860$ $[+0.0566, +0.1154]$	3.47×10^{-8}	1.04×10^{-7}
Poem	Llama	$+0.0960$ $[+0.0830, +0.1090]$	4.62×10^{-32}	4.16×10^{-31}
Poem	Qwen	$+0.0380$ $[+0.0278, +0.0482]$	5.03×10^{-12}	3.02×10^{-11}
Poem	Gemma	$+0.0440$ $[+0.0311, +0.0569]$	2.03×10^{-10}	1.01×10^{-9}
Story	Llama	$+0.0920$ $[+0.0695, +0.1145]$	6.31×10^{-14}	4.41×10^{-13}
Story	Qwen	$+0.0460$ $[+0.0263, +0.0657]$	7.13×10^{-6}	1.43×10^{-5}
Story	Gemma	$+0.0700$ $[+0.0472, +0.0928]$	7.66×10^{-9}	3.06×10^{-8}
Diversity: SLR − Proxy-Interp.				
Joke	Llama	$+0.0330$ $[-0.0023, +0.0683]$	0.0666	0.0666
Joke	Qwen	-0.0300 $[-0.0441, -0.0159]$	4.12×10^{-5}	2.47×10^{-4}
Joke	Gemma	$+0.1040$ $[+0.0849, +0.1231]$	1.48×10^{-20}	1.18×10^{-19}
Poem	Llama	$+0.0870$ $[+0.0737, +0.1003]$	1.29×10^{-27}	1.16×10^{-26}
Poem	Qwen	$+0.0110$ $[+0.0011, +0.0209]$	0.0300	0.0601
Poem	Gemma	$+0.0280$ $[+0.0146, +0.0414]$	5.67×10^{-5}	2.83×10^{-4}
Story	Llama	$+0.0690$ $[+0.0463, +0.0917]$	1.01×10^{-8}	7.08×10^{-8}
Story	Qwen	$+0.0310$ $[+0.0109, +0.0511]$	0.0027	0.0080
Story	Gemma	$+0.0380$ $[+0.0149, +0.0611]$	0.0014	0.0056

Table 22: Open-ended QA differences between SLR and each baseline at $T = 1$. Pointwise 95% confidence intervals and raw two-sided p-values use Welch inference over 40 prompts. Holm-adjusted p-values are also reported. Differences are in the original metric units.

Model	Metric	Δ	95% CI	p_{raw}	p_{Holm}
SLR − Post-trained					
Llama	Precision	-0.0180	$[-0.0467, +0.0107]$	0.2152	0.4304
Llama	Entropy	$+0.8800$	$[+0.5771, +1.1829]$	1.45×10^{-7}	5.80×10^{-7}
Llama	Coverage-n	$+0.2570$	$[+0.1647, +0.3493]$	6.22×10^{-7}	1.24×10^{-6}
Qwen	Precision	-0.0210	$[-0.0463, +0.0043]$	0.1008	0.3024
Qwen	Entropy	$+0.9240$	$[+0.6597, +1.1883]$	1.05×10^{-9}	6.30×10^{-9}
Qwen	Coverage-n	$+0.1400$	$[+0.0828, +0.1972]$	8.29×10^{-6}	8.29×10^{-6}
Gemma	Precision	-0.0160	$[-0.0477, +0.0157]$	0.3154	0.4304
Gemma	Entropy	$+1.0560$	$[+0.7541, +1.3579]$	1.11×10^{-9}	6.30×10^{-9}
Gemma	Coverage-n	$+0.2110$	$[+0.1398, +0.2822]$	2.51×10^{-7}	7.53×10^{-7}
SLR − Proxy-Interp.					
Llama	Precision	-0.0130	$[-0.0430, +0.0170]$	0.3909	1.0000
Llama	Entropy	$+0.5620$	$[+0.2499, +0.8741]$	5.87×10^{-4}	0.0018
Llama	Coverage-n	$+0.1760$	$[+0.0770, +0.2750]$	7.05×10^{-4}	0.0018
Qwen	Precision	$+0.0010$	$[-0.0360, +0.0380]$	0.9572	1.0000
Qwen	Entropy	$+0.6640$	$[+0.3789, +0.9491]$	1.40×10^{-5}	5.60×10^{-5}
Qwen	Coverage-n	$+0.1020$	$[+0.0392, +0.1648]$	0.0018	0.0018
Gemma	Precision	-0.0150	$[-0.0472, +0.0172]$	0.3530	1.0000
Gemma	Entropy	$+0.8600$	$[+0.6171, +1.1029]$	1.01×10^{-8}	6.06×10^{-8}
Gemma	Coverage-n	$+0.1860$	$[+0.1134, +0.2586]$	3.97×10^{-6}	1.98×10^{-5}

GSM8K. We use two-sided question-paired sign-flip tests for Pass@1–32 and exact McNemar/binomial tests for Pass@64 on the model-family-specific subsets of 100 questions in Table 3. Each question has 64 samples, scored using the exact-match protocol in Appendix B.1. The 18 model–budget comparisons against Post-trained form one Holm correction family. All raw p-values are at most 1.53×10^{-5}, so their Holm-adjusted values are at most $18 \times 1.53 \times 10^{-5} = 2.754 \times 10^{-4}$; all 18 gains remain significant at the 0.05 level. SLR also achieves higher Pass@k than Proxy-Interpolation on all 18 comparisons.

C.13 Comparison with pre-trained distributions

We compare the pre-trained, post-trained, and SLR checkpoints on 10 digit CRC prompts using a shared completion interface. Each checkpoint receives the same prefix, `Q: {instruction}\nA:`, followed by one ASCII space; `\n` denotes a newline. We apply no chat template or additional special tokens. At the first generated position, we measure the probability assigned to valid digit tokens and the entropy of their renormalized distribution, as defined in Section 3.2. We compare each post-trained and SLR distribution with its pre-trained counterpart using Jensen–Shannon (JS) divergence and total variation (TV) distance over the valid digits. Table 23 reports means over prompts.

Table 23: Controlled digit-completion diagnostic. Pre and Post denote the pre-trained and post-trained checkpoints. Entropy and JS use natural logarithms. Distances compare the normalized valid-digit distributions with Pre; lower values indicate closer distributions.

Model	Valid mass Pre / Post / SLR	Entropy (nats) Pre / Post / SLR	JS to Pre Post / SLR	TV to Pre Post / SLR
Llama	0.934 / 0.961 / 0.962	1.744 / 1.395 / 1.707	0.068 / 0.011	0.275 / 0.112
Qwen	0.996 / 1.000 / 0.999	1.640 / 0.474 / 1.311	0.218 / 0.034	0.529 / 0.222
Gemma	0.491 / 0.291 / 0.309	1.453 / 1.345 / 1.550	0.102 / 0.089	0.370 / 0.345

For Llama and Qwen, SLR is closer to the pre-trained distribution under both distances on every digit prompt, while valid-token probability remains high. Gemma assigns less probability to valid digits under the shared interface, so its conditional distances provide weaker evidence of distributional recovery.

D Existing assets, licenses, and terms of use

We use only existing publicly available or API-accessible assets, and use them solely for research evaluation. We do not redistribute any third-party model weights, proprietary API models, or external datasets. The CRC prompts introduced in this paper are synthetic digit/letter constraint prompts and do not contain personal or sensitive information. The existing assets used in this work are as follows.

- **Llama models.** We use Llama-3.1-8B and Llama-3.1-8B-Instruct [8] as a pre-trained/post-trained checkpoint pair for SLR, Proxy-Interpolation, and downstream evaluation. These models are governed by the Llama 3.1 Community License and Acceptable Use Policy.
- **Qwen models.** We use Qwen2.5-7B / Qwen2.5-7B-Instruct and Qwen2.5-32B / Qwen2.5-32B-Instruct [34] for the main and larger-model experiments. These models are released under the Apache-2.0 license.
- **Gemma models.** We use Gemma2-9B / Gemma2-9B-It and Gemma2-27B / Gemma2-27B-It [7] for the main and larger-model experiments. These models are governed by the Gemma Terms of Use.
- **Embedding and judge models.** We use Qwen3-Embedding-8B [39], released under Apache-2.0, to compute semantic diversity in creative-writing experiments. We use Gemini-2.5-Pro [4] as an API-based LLM judge for creative-writing quality and as a canonicalizer for Open-ended QA correctness, under the applicable Google AI Studio / Gemini API terms.
- **Creative-writing and Open-ended QA datasets.** We use the creative-writing prompts and verbalized sampling prompt template from Verbalized Sampling [38]; the code repository is released under Apache-2.0, while dataset assets follow their released dataset terms. We also use the Verbalized Sampling Open-ended QA dataset [38], which is based on CoverageQA/SimpleStrat [31]; this dataset is governed by the Verbalized Sampling dataset agreement and requires attribution. The original SimpleStrat/CoverageQA repository currently does not specify a standard license.
- **Reasoning, instruction-following, and safety datasets.** We use GSM8K [3] for reasoning evaluation, which is released under the MIT License. We use IFEval [40] for instruction-following evaluation, released under Apache-2.0. We use XSTest [22] for safety-related refusal evaluation; its test prompts are released under CC-BY-4.0, while generated completions are subject to the licenses or terms of the evaluated models.
- **HMA baseline.** We use the official HMA / alignment-tax implementation from Lin et al. [15] as an additional weight-space baseline for Open-ended QA. The code is released under Apache-2.0, with additional repository terms applying where relevant.

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